Practical 1

Aim : Installation and setup of java development kit(JDK),setup android SDK,setup eclipse IDE,setup android development tools (ADT) plugins,create android virtual device.

Environment setup for Android apps Development

To setup an environment for the Android application development, you will need to download and install the following:

- Java Development Kit (JDK) 5 or 6
- The Android SDK
- Eclipse

Step 1: Download and Install JDK

The Android SDK makes use of the Java SE Development Kit (JDK). If your computer does not have the JDK installed, you should start by downloading it from: **www.oracle.com/technetwork/java/javase/downloads/index.html** and installing it prior to moving to the next section.

Figure 1

Step 2: Download Android Development Environment

There are a number of ways to develop Android apps. The easiest way to get started quickly is to use the ADT Bundle developed by Google. The ADT Bundle provides a number of tools that are essential to building android apps.

To download ADT bundle go to https://developer.android.com/sdk/index.html. On this web page click on the button "Download the SDK" as shown in below figure.

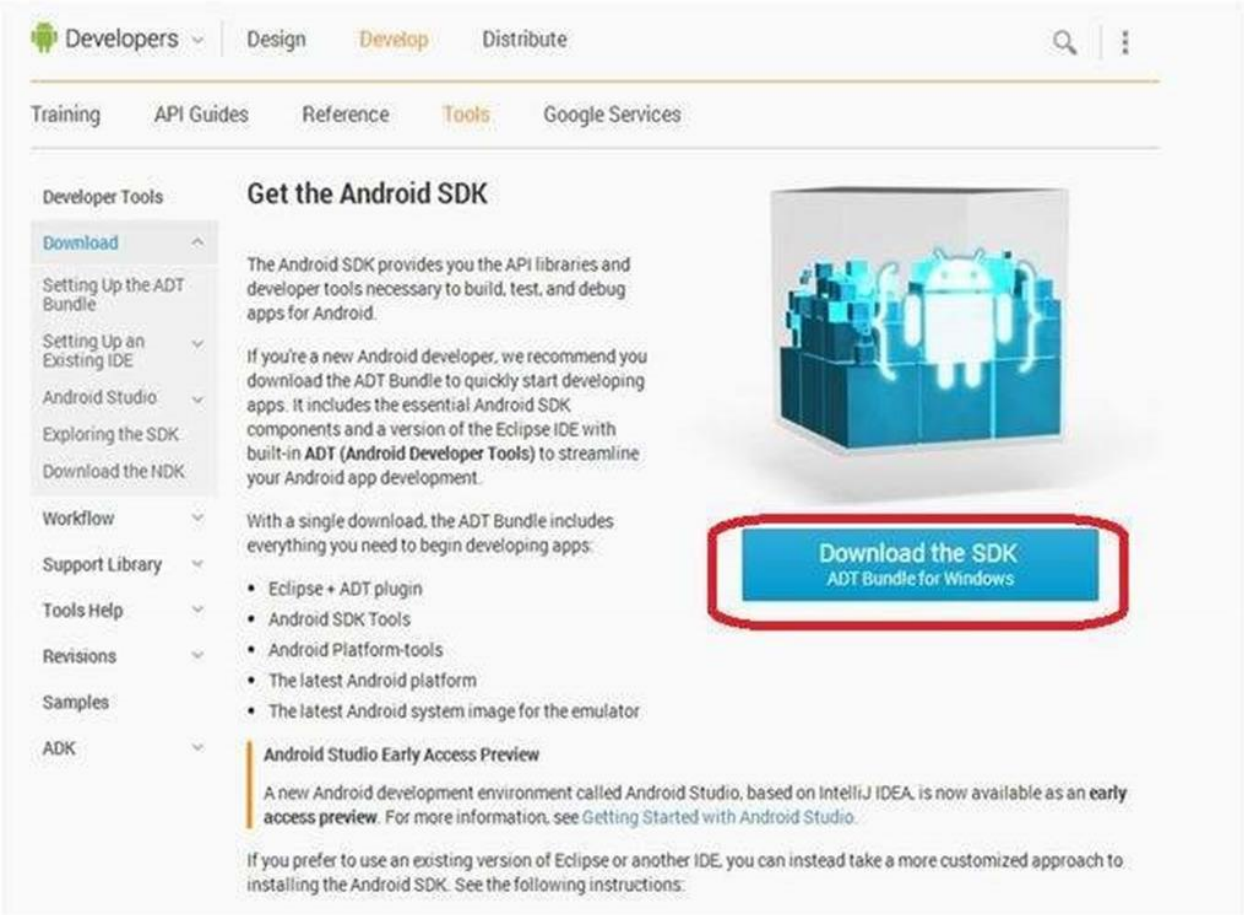


Figure 2

When you click on this button, it will ask to agree on terms and condition, now click the checkbox agreeing to the terms and conditions and select either 32-bit or 64-bit, depending on your computer. After you have agreed and selected your system type click on the "Download the SDK ADT Bundle for Windows" button and the download will start. The downloaded file is a .zip file

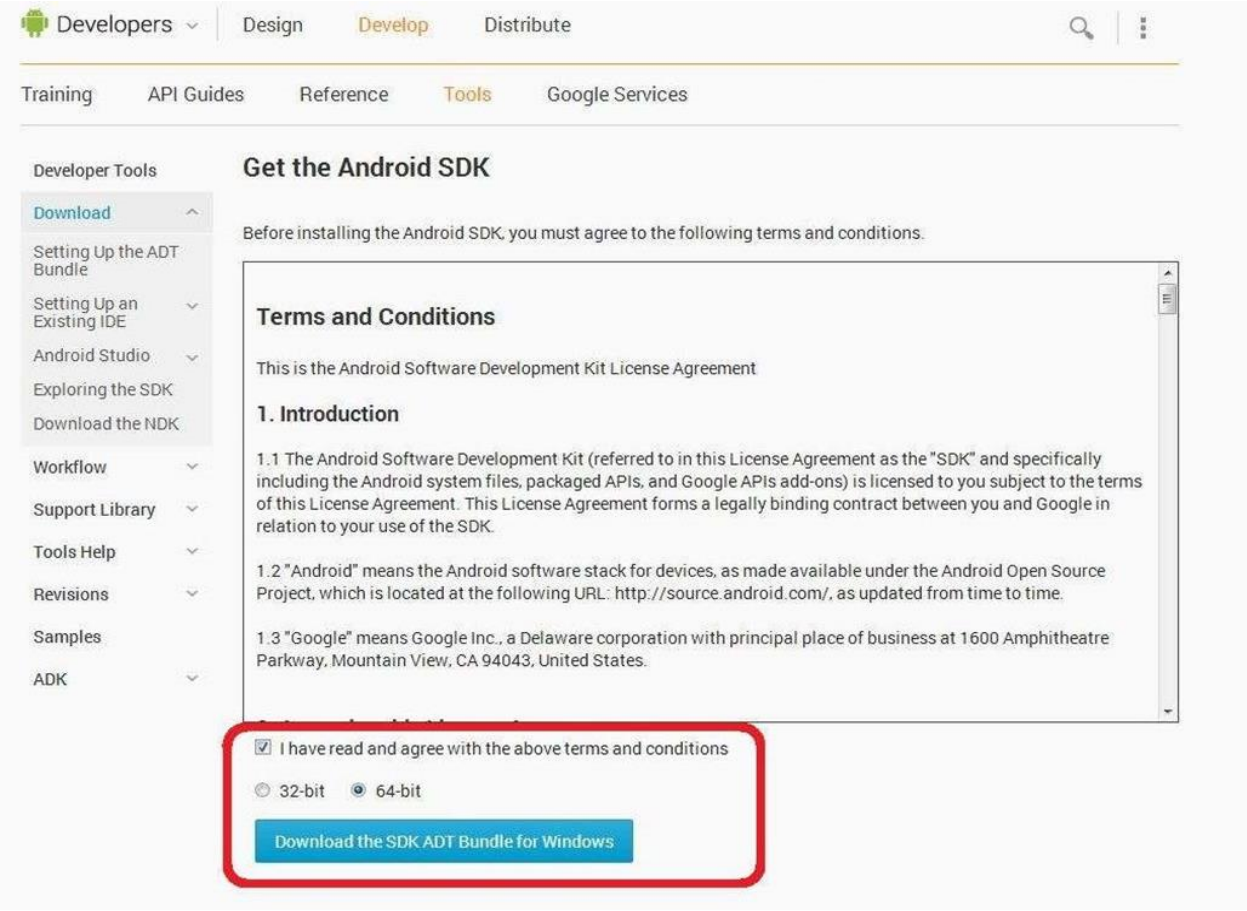


Figure 3

Step 3: Installing the ADT bundle

Once downloading of the ADT Bundle has completed, Extract all the files to a location on your computer. You will find two folders named eclipse and sdk and an application called SDK Manager as shown in below figure.

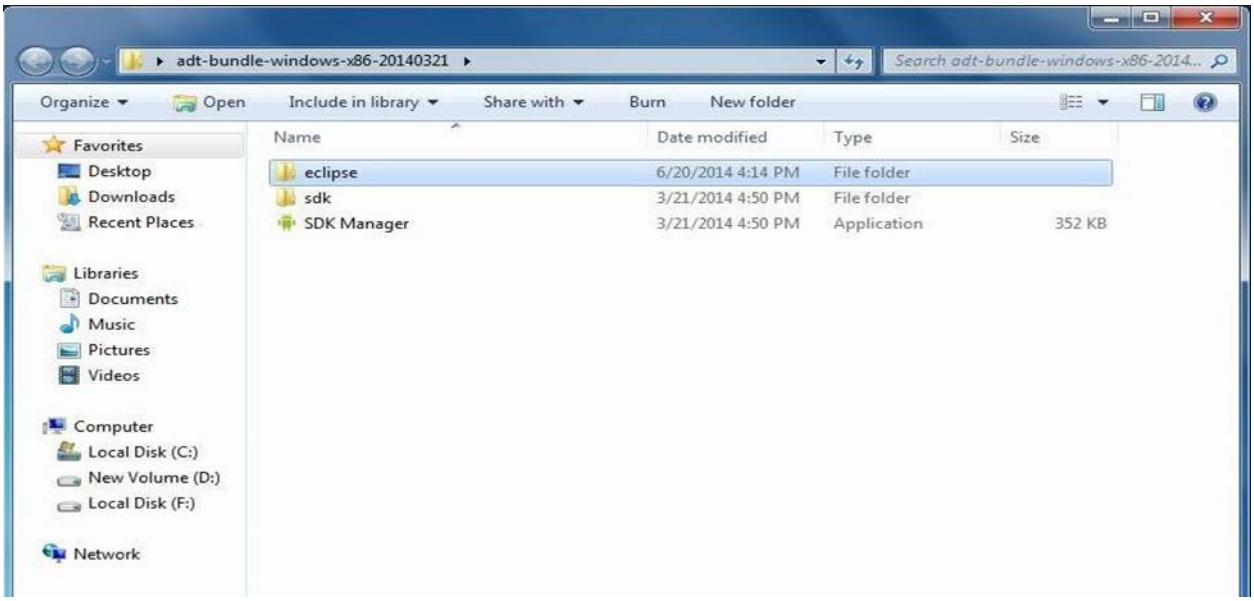


Figure 4

Now you are going to open up the environment where you will be developing android applications by click on eclipse folder. You will now see multiple folders and files as sown in below figure. You need to double click on eclipse.exe to open the eclipse. Once eclipses get started, you can start building your first application.

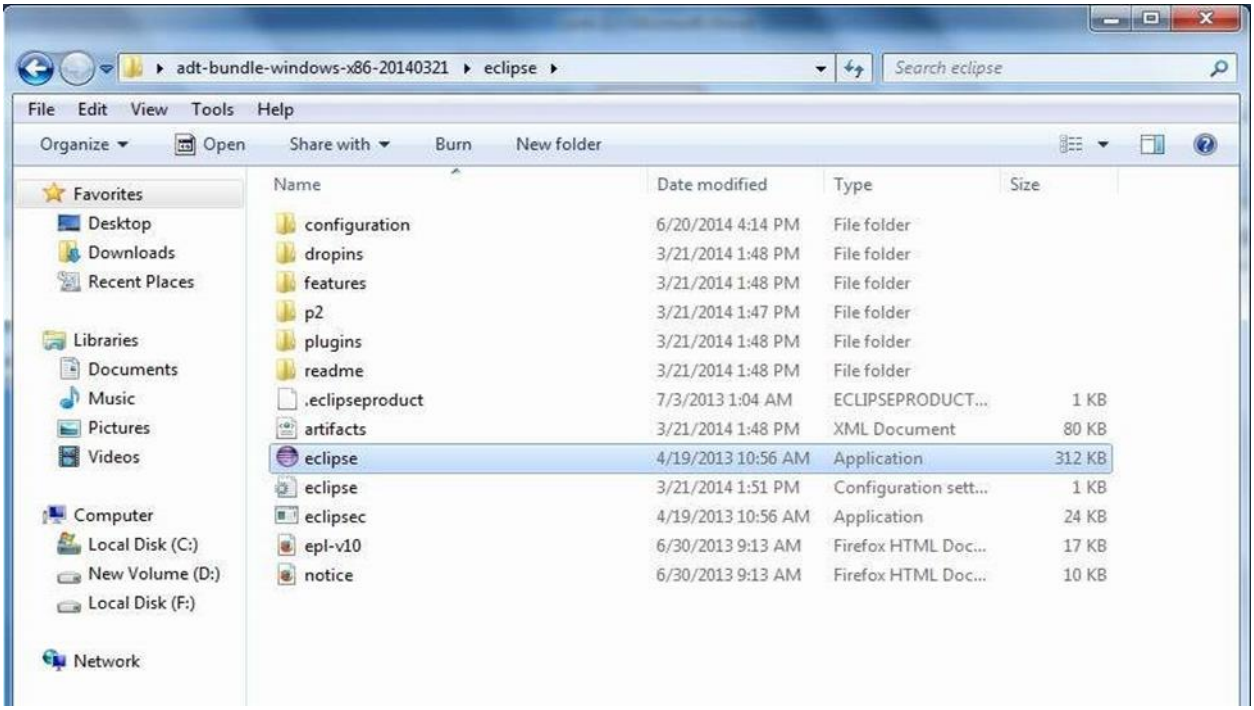


Figure 5

Practical 2

Aim : Create “Hello World” application. That will display “Hello World” in the middle of the screen using TextView Widget in the red color.

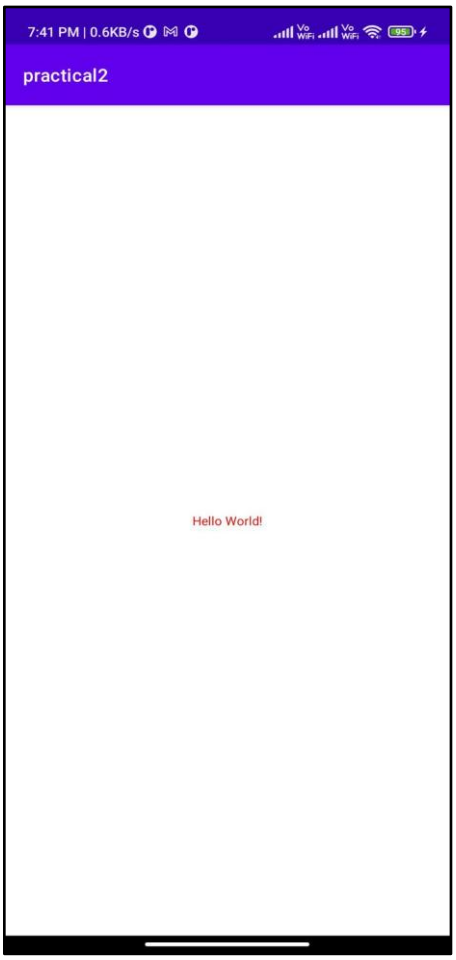
Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".MainActivity">


    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Hello World!"

        android:textColor="#FF0000"

        app:layout_constraintBottom_toBottomOf="parent"

        app:layout_constraintLeft_toLeftOf="parent"

        app:layout_constraintRight_toRightOf="parent"

        app:layout_constraintTop_toTopOf="parent" />


</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical1;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output



Practical 3

Aim : Create application for demonstration of android activity life cycle.

Activity_main.xml

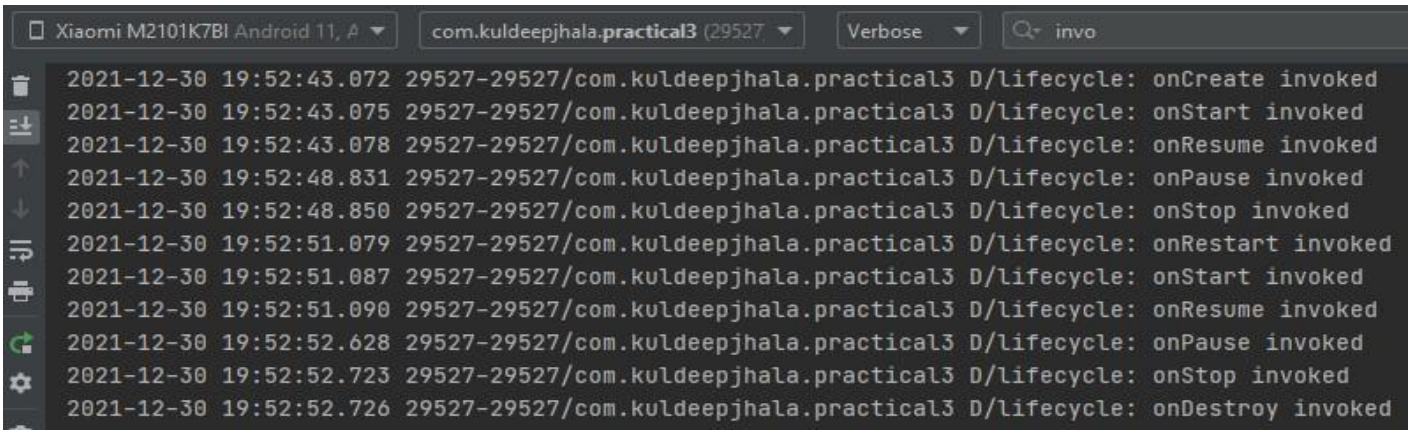
```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".MainActivity">


    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Hello World!"

        app:layout_constraintBottom_toBottomOf="parent"

        app:layout_constraintLeft_toLeftOf="parent"

        app:layout_constraintRight_toRightOf="parent"

        app:layout_constraintTop_toTopOf="parent" />


</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical3;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Log.d("lifecycle","onCreate invoked");
    }
    @Override
    protected void onStart()
    {
        super.onStart();
        Log.d("lifecycle", "onStart invoked");
    }
    @Override
    protected void onResume() {
        super.onResume();
        Log.d("lifecycle","onResume invoked");
    }
    @Override
    protected void onPause() {
        super.onPause();
        Log.d("lifecycle","onPause invoked");
    }
```

```
    @Override
    protected void onStop() {
        super.onStop();
        Log.d("lifecycle","onStop invoked");
    }
    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d("lifecycle","onRestart invoked");
    }
    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d("lifecycle","onDestroy invoked");
    }
}
```

Output



Practical 4

Aim : Create Registration page to demonstration of Basic widgets available in android.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:layout_margin="20dp"

    android:background="#2D272A2B"

    tools:context=".MainActivity">

    <EditText

        android:id="@+id/edit_text_first_name"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:layout_marginBottom="30dp"

        android:hint="Enter first Name"

        app:layout_constraintBottom_toTopOf="@id/edit_text_last_name"

        app:layout_constraintEnd_toEndOf="parent"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintTop_toTopOf="parent" />

    <EditText
```

```
        android:id="@+id/edit_text_last_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:layout_marginBottom="30dp"
        android:hint="Enter last Name"
        app:layout_constraintBottom_toTopOf="@id/edit_text_address"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/edit_text_first_name" />
    <EditText
        android:id="@+id/edit_text_address"
        android:layout_width="378dp"
        android:layout_height="100dp"
        android:layout_marginBottom="30dp"
        android:hint="Enter our address"
        app:layout_constraintBottom_toTopOf="@id/edit_text_phoneNo"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/edit_text_last_name" />
    <EditText
        android:id="@+id/edit_text_phoneNo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

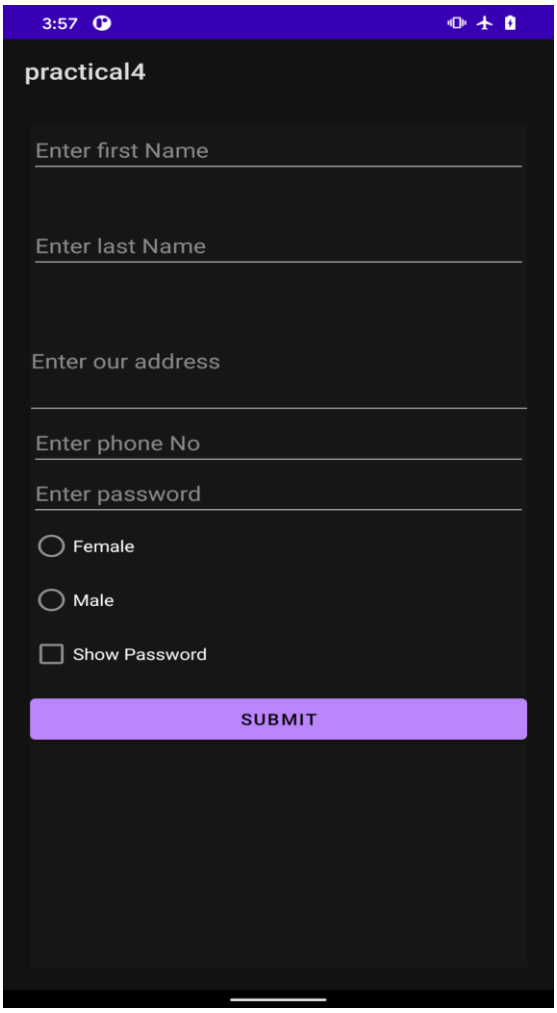
```
        android:layout_marginBottom="30dp"
        android:hint="Enter phone No"
        android:inputType="phone"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/edit_text_address" />
    <EditText
        android:id="@+id/edit_text_password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="30dp"
        android:hint="Enter password"
        android:inputType="phone"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/edit_text_phoneNo" />
    <CheckBox
        android:id="@+id/check_box_show_password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="30dp"
        android:text="Show Password"
        app:layout_constraintTop_toBottomOf="@id/radio_ninjas" />
    <RadioButton
```

```
        android:id="@+id/radio_pirates"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female"
        app:layout_constraintBottom_toTopOf="@id/radio_ninjas"
        app:layout_constraintTop_toBottomOf="@id/edit_text_password"
        tools:ignore="MissingConstraints" />
    <RadioButton
        android:id="@+id/radio_ninjas"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male"
        app:layout_constraintBottom_toTopOf="@id/check_box_show_password"
        app:layout_constraintTop_toBottomOf="@id/radio_pirates"
        tools:ignore="MissingConstraints" />
    <Button
        android:id="@+id/btn_submit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="Submit"
        app:layout_constraintTop_toBottomOf="@id/check_box_show_password"
        tools:layout_editor_absoluteX="-3dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practicals;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class p4_MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.p4_activity_main);
    }
}
```

Output



Practical 5

Aim : Create sample application with login module.(Check username and password) On successful login, Change TextView “Login Successful”. And on failing login, alert user using Toast “Login fail”

Activity_main.xml

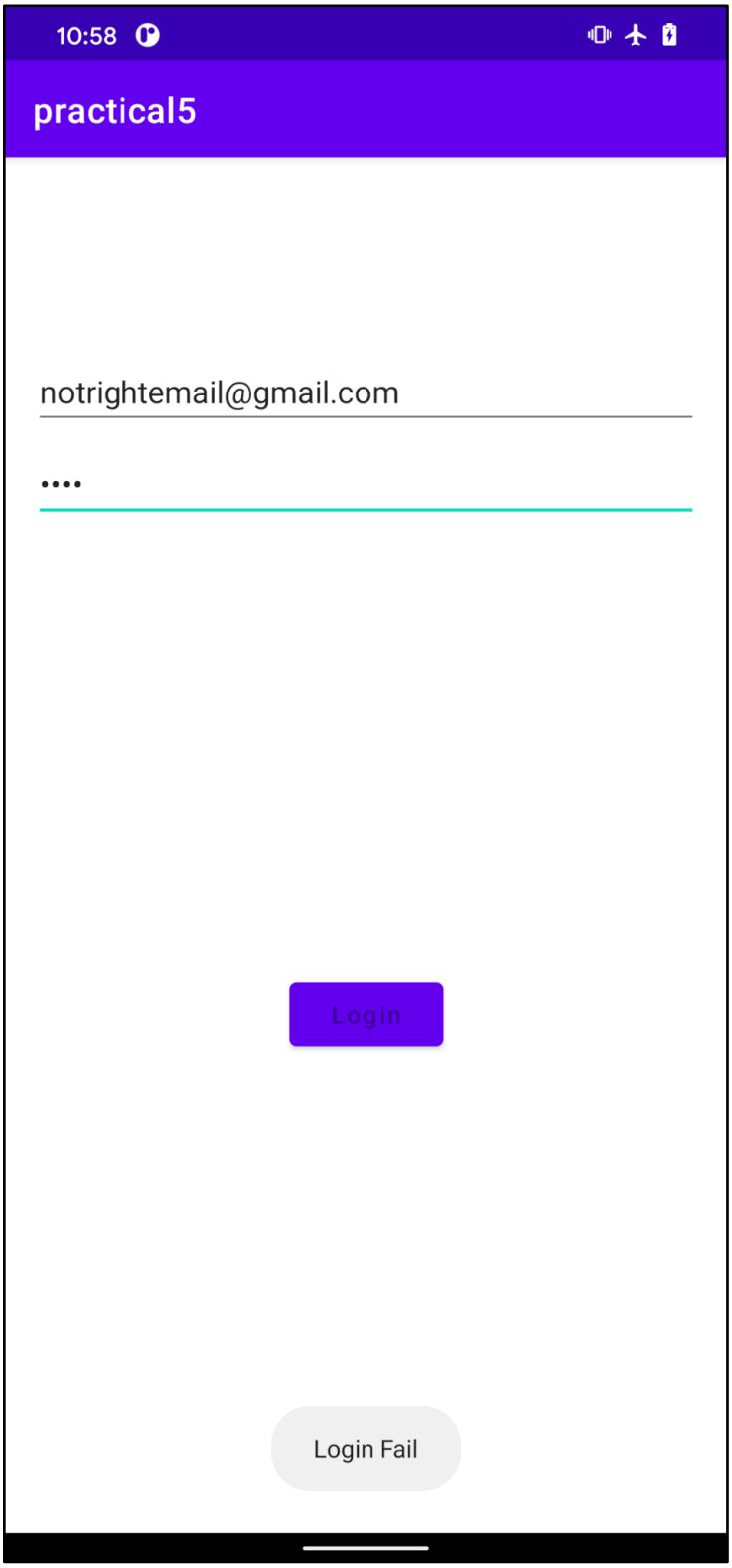
```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:paddingLeft="16dp"

    android:paddingTop="16dp"

    android:paddingRight="16dp"

    android:paddingBottom="16dp"

    tools:context=".MainActivity">


    <EditText

        android:id="@+id/username"

        android:layout_width="0dp"

        android:layout_height="wrap_content"

        android:layout_marginTop="96dp"

        android:hint="Email"

        android:inputType="textEmailAddress"

        android:selectAllOnFocus="true"

        app:layout_constraintEnd_toEndOf="parent"

        app:layout_constraintStart_toStartOf="parent"
```

```
        app:layout_constraintTop_toTopOf="parent" />

    <EditText
        android:id="@+id/password"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:hint="Password"
        android:imeOptions="actionDone"
        android:inputType="textPassword"
        android:selectAllOnFocus="true"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/username" />

    <Button
        android:id="@+id/login"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:hint="Login"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/password" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical5;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.EditText;
import android.widget.Toast;
import com.kuldeepjhala.practical5.databinding.ActivityMainBinding;
public class MainActivity extends AppCompatActivity {
ActivityMainBinding activityMainBinding;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        activityMainBinding = ActivityMainBinding.inflate(getLayoutInflater());
        setContentView(activityMainBinding.getRoot());
        EditText usernameEditText = activityMainBinding.username;
        EditText passwordEditText = activityMainBinding.password;
        String emailid = "123@gmail.com";
        String password = "123";
        activityMainBinding.login.setOnClickListener(l -> {
                if (emailid.equals(usernameEditText) && password.equals(passwordEditText)) {
                    Toast.makeText(getApplicationContext(), "Login Successfull",
Toast.LENGTH_LONG).show();
                } else {
                    Toast.makeText(getApplicationContext(), "Login Fail",
Toast.LENGTH_SHORT).show();
                }
            }
        );
    }
}
```

Output



Practical 6

Aim : Create sample application with login module.(Check username and password) On successful login, Change TextView “Login Successful”. And on failing login, alert user using Toast “Login fail”

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:paddingLeft="16dp"

    android:paddingTop="16dp"

    android:paddingRight="16dp"

    android:paddingBottom="16dp"

    tools:context=".MainActivity">


    <EditText

        android:id="@+id/username"

        android:layout_width="0dp"

        android:layout_height="wrap_content"

        android:layout_marginTop="96dp"

        android:hint="Email"

        android:inputType="textEmailAddress"

        android:selectAllOnFocus="true"

        app:layout_constraintEnd_toEndOf="parent"

        app:layout_constraintStart_toStartOf="parent"
```

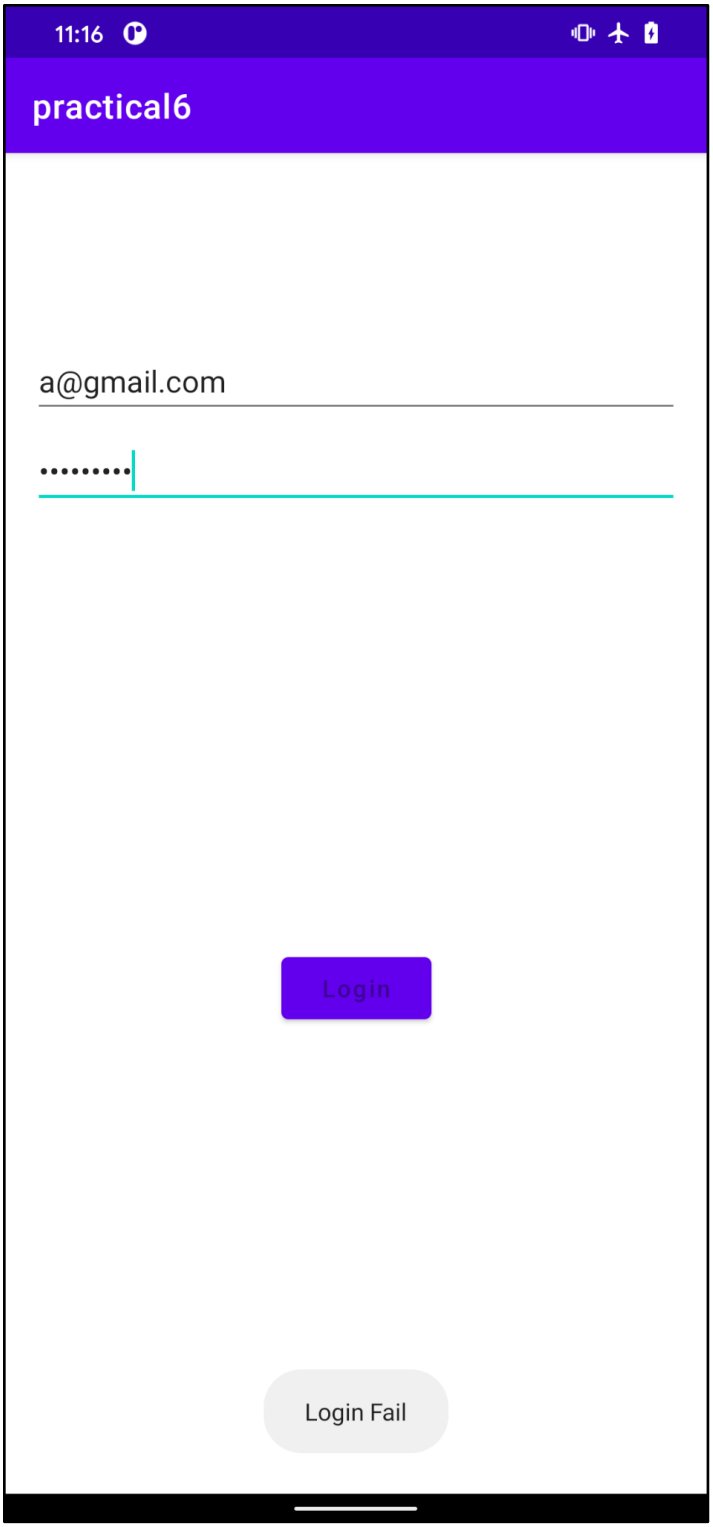
```
        app:layout_constraintTop_toTopOf="parent" />

    <EditText
        android:id="@+id/password"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:hint="Password"
        android:imeOptions="actionDone"
        android:inputType="textPassword"
        android:selectAllOnFocus="true"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/username" />

    <Button
        android:id="@+id/login"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:hint="Login"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/password" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical6;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.widget.EditText;

import android.widget.Toast;

import com.kuldeepjhala.practical6.databinding.ActivityMainBinding;

public class MainActivity extends AppCompatActivity {

    ActivityMainBinding activityMainBinding;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        activityMainBinding = ActivityMainBinding.inflate(getLayoutInflater());

        setContentView(activityMainBinding.getRoot());

        EditText usernameEditText = activityMainBinding.username;

        EditText passwordEditText = activityMainBinding.password;

        String emailid = "123@gmail.com";

        String password = "123";

        activityMainBinding.login.setOnClickListener(l -> {

                if (emailid.equals(usernameEditText.toString()) &&
password.equals(passwordEditText.toString())) {

                    Toast.makeText(getApplicationContext(), "Login Successfull",
Toast.LENGTH_LONG).show();

                } else {

                    Toast.makeText(getApplicationContext(), "Login Fail",
Toast.LENGTH_SHORT).show();

                }

            }

        );

    }
```

}

Output



Practical 7

Aim : Create and Login application as above. Validate login data and display Error to user using setError() method.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="?android:attr/colorButtonNormal"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="135dp"
        android:layout_height="34dp"
        android:layout_x="2dp"
        android:layout_y="276dp"
        android:text="EMAIL"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintHorizontal_bias="0.105"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.315" />
    <TextView
        android:id="@+id/textView"
        android:layout_width="121dp"
```

```
        android:layout_height="44dp"
        android:layout_x="6dp"
        android:layout_y="354dp"
        android:text="PASSWORD"
        android:textSize="22sp"
        tools:layout_editor_absoluteX="43dp"
        tools:layout_editor_absoluteY="306dp" />
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="153dp"
        android:layout_y="450dp"
        android:text="LOGIN"
        tools:layout_editor_absoluteX="164dp"/>
    <EditText
        android:id="@+id/editText1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="146dp"
        android:layout_y="271dp"
        android:ems="10"
        android:inputType="textPersonName"
        android:text="Name"
        tools:layout_editor_absoluteX="173dp"
        tools:layout_editor_absoluteY="209dp" />
    <EditText
        android:id="@+id/editText2"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="146dp"
        android:layout_y="351dp"
        android:ems="10"
        android:inputType="textPersonName"
        android:text="Name"
        tools:layout_editor_absoluteX="178dp"
        tools:layout_editor_absoluteY="305dp" />
    <TextView
        android:id="@+id/textView2"
        android:layout_width="182dp"
        android:layout_height="50dp"
        android:layout_x="106dp"
        android:layout_y="126dp"
        android:text="LOGIN PAGE"
        android:textColor="@android:color/black"
        android:textSize="30sp"
        tools:layout_editor_absoluteX="114dp"
        tools:layout_editor_absoluteY="69dp" />
</AbsoluteLayout>
```

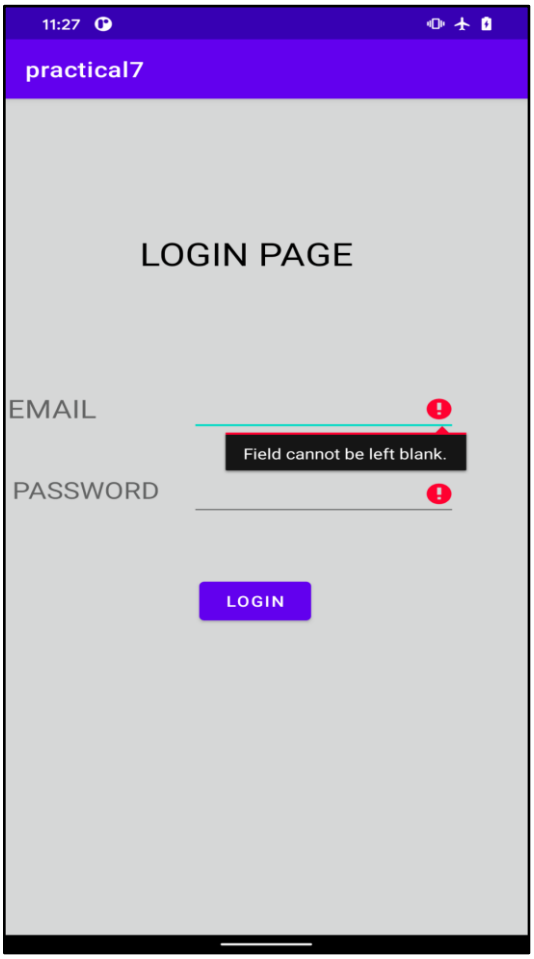
MainActivity.java

```
package com.kuldeepjhala.practical7;
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import java.util.regex.Matcher;
import java.util.regex.Pattern;
public class MainActivity extends AppCompatActivity {
    EditText et1,et2;
    Button b1;
    Pattern p1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        et1 = findViewById(R.id.editText1);
        et2 = findViewById(R.id.editText2);
        b1 = findViewById(R.id.button1);
        p1 = Pattern.compile(".+@.+\\.[a-z]+");
        et1.setOnFocusChangeListener(new View.OnFocusChangeListener() {
            @Override
            public void onFocusChange(View v, boolean hasFocus) {
                // TODO Auto-generated method stub
                String s1 = et1.getText().toString();
                Matcher m = p1.matcher(s1);
                if(et1.getText().length()==0){
                    if(et2.getText().length()==0)
                    {
                        et1.setError("Field cannot be left blank.");
```

```
                et2.setError("Field cannot be left blank.");
            }
        }
    }
    });
  }
}
```

Output



Practical 8

Aim : Create an application for demonstration of Relative and Table Layout in android.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    android:background="@color/bg"

    android:layout_height="wrap_content">

    <RelativeLayout

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:id="@+id/relative_layout"

        tools:ignore="MissingConstraints">

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:layout_marginTop="10dp"

        android:layout_centerHorizontal="true"

        android:text="relative layout one"

        android:textSize="20sp"/>

    <TextView

        android:layout_width="wrap_content"

        android:layout_centerHorizontal="true"

        android:layout_marginTop="50dp"
```

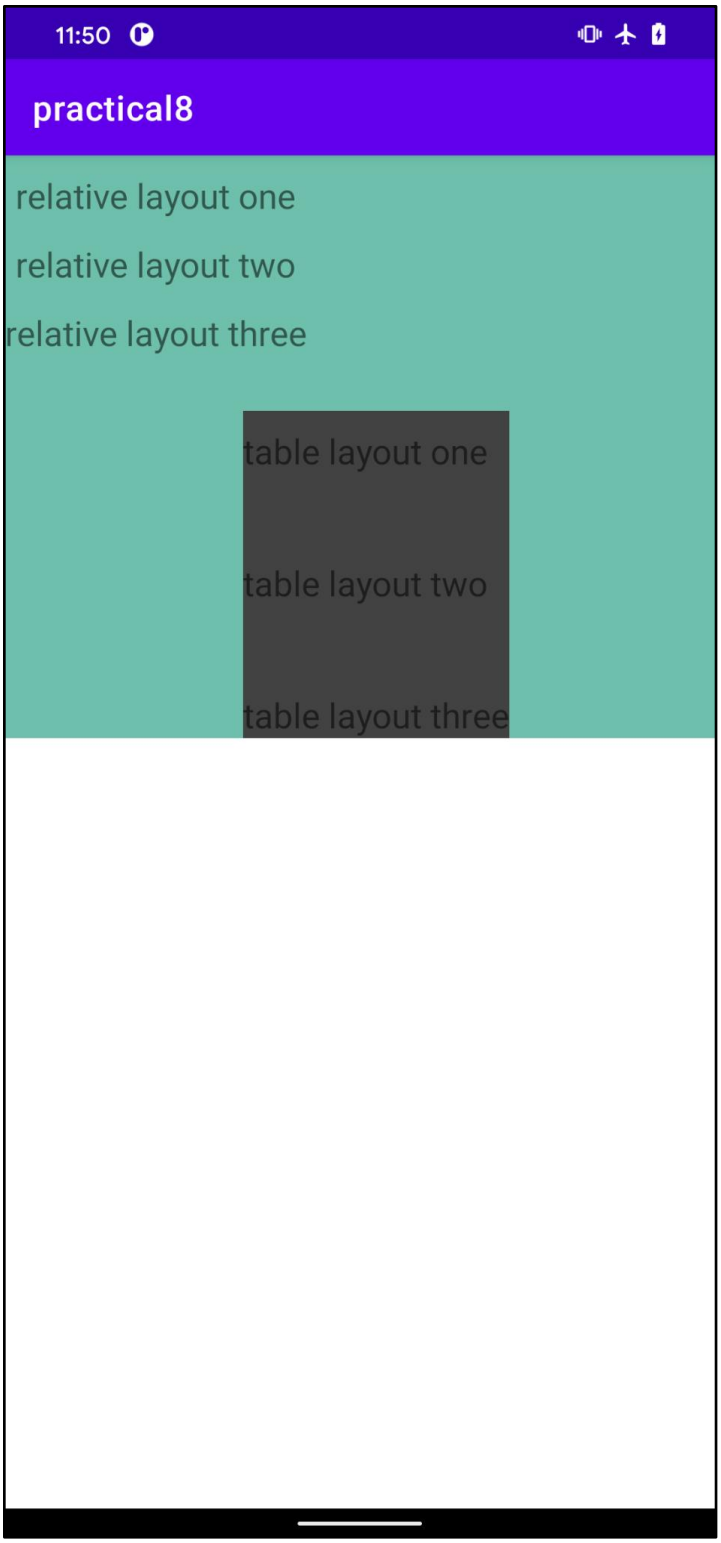
```
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:text="relative layout two" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="90dp"
        android:layout_centerHorizontal="true"
        android:textSize="20sp"
        android:text="relative layout three" />
    </RelativeLayout>
    <TableLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:background="@color/cardview_dark_background"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.536"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/relative_layout">
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_centerHorizontal="true"
            android:layout_marginTop="10dp"
            android:text="table layout one"
            android:textSize="20sp" />
        <TextView
```

```
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_centerHorizontal="true"
            android:layout_marginTop="50dp"
            android:text="table layout two"
            android:textSize="20sp" />
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_centerHorizontal="true"
            android:layout_marginTop="50dp"
            android:text="table layout three"
            android:textSize="20sp" />
    </TableLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical8;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output


11:50
practical8
relative layout one
relative layout two
relative layout three
table layout one
table layout two
table layout three

Practical 9

Aim : Create an application for demonstration of Scroll view in android

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:id="@+id/constraint"

    android:layout_height="match_parent"

    tools:context=".MainActivity">

    <ScrollView

        android:layout_width="409dp"

        android:layout_height="729dp"

        tools:layout_editor_absoluteX="1dp"

        tools:layout_editor_absoluteY="1dp"

        tools:ignore="MissingConstraints">

        <LinearLayout

            android:layout_width="match_parent"

            android:layout_height="wrap_content"

            android:orientation="vertical">

        <TextView

            android:layout_width="wrap_content"

            android:layout_height="wrap_content"

            android:text="Hello World!"

            app:layout_constraintBottom_toBottomOf="parent"

            app:layout_constraintLeft_toLeftOf="parent"
```

```
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="100dp"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="100dp"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="100dp"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
```

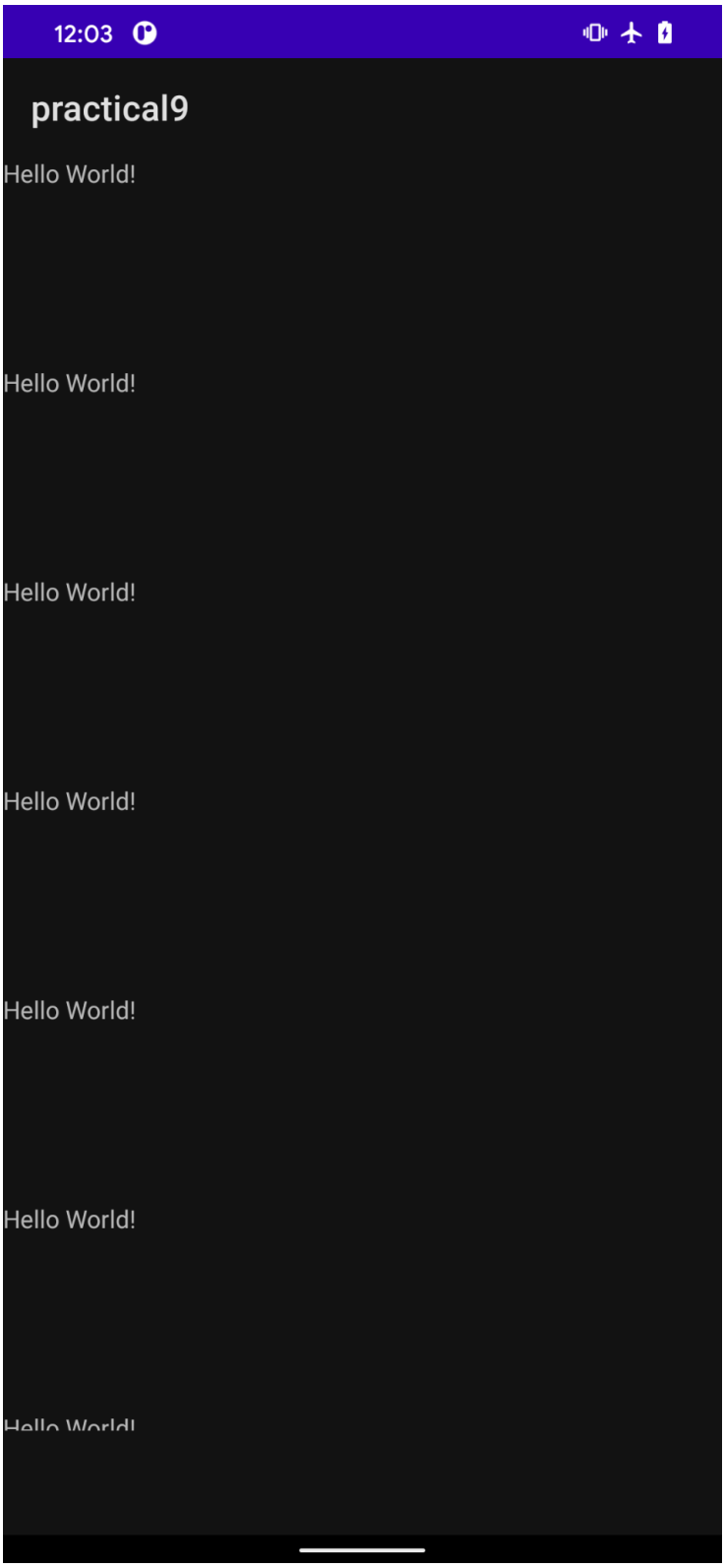
```
        app:layout_constraintTop_toTopOf="parent" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="100dp"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Hello World!"
            android:layout_marginTop="100dp"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintLeft_toLeftOf="parent"
            app:layout_constraintRight_toRightOf="parent"
            app:layout_constraintTop_toTopOf="parent" />
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="100dp"
            android:text="Hello World!"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintLeft_toLeftOf="parent"
            app:layout_constraintRight_toRightOf="parent"
            app:layout_constraintTop_toTopOf="parent" />
```

```
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="50dp"
            android:text="Hello World!"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintLeft_toLeftOf="parent"
            app:layout_constraint Right_toRightOf="parent"
            app:layout_constraintTop_toTopOf="parent" />
    </LinearLayout>
  </ScrollView>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical9;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
  }
```

Output


12:03
practical9
Hello World!
Hello World!
Hello World!
Hello World!
Hello World!
Hello World!
Hello World!

Practical 10

Aim : Create an application for demonstration of Explicitly Starting New Activity using Intent.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".MainActivity">

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="ActivityOne"

        android:textSize="20sp"

        android:typeface="serif"

        app:layout_constraintTop_toTopOf="parent"

        app:layout_constraintTop_toBottomOf="parent"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintEnd_toEndOf="parent"

        />

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:id="@+id/click"

        android:text="Click me"
```

```
        android:textSize="30dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="ActivityTwo"
        android:textSize="20sp"
        android:typeface="serif"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintTop_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical10;
```

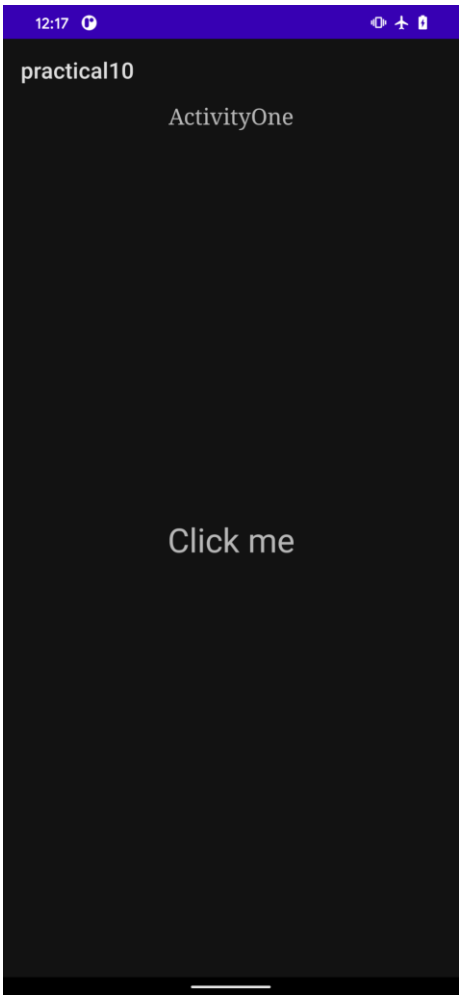
```
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import com.kuldeepjhala.practical10.databinding.ActivityMainBinding;
public class MainActivity extends AppCompatActivity {
ActivityMainBinding activityMainBinding;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        activityMainBinding = ActivityMainBinding.inflate(getLayoutInflater());
        setContentView(activityMainBinding.getRoot());
        activityMainBinding.click.setOnClickListener( l->{
            Intent i = new Intent(getApplicationContext(), MainActivity2.class);
            startActivity(i);
                }
        );
    }
}
```

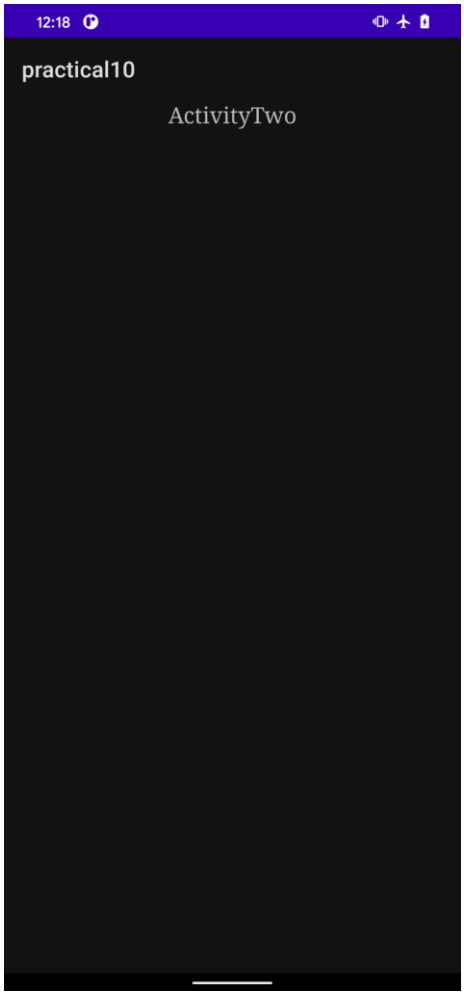
MainActivity2.java

```
package com.kuldeepjhala.practical10;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
    }
```

}

Output


12:17
practical10
ActivityOne
Click me


12:18
practical10
ActivityTwo

Practical 11

Aim : Create an application that will pass two number using TextView to the next screen , and on the next screen display sum of that number.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".MainActivity">

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="ActivityOne"

        android:textSize="20sp"

        android:typeface="serif"

        app:layout_constraintTop_toTopOf="parent"

        app:layout_constraintTop_toBottomOf="parent"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintEnd_toEndOf="parent"

        />

    <TextView

        android:id="@+id/tv_1"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="10"
```

```
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintHorizontal_bias="0.215"
        android:textSize="30sp"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.247" />
    <TextView
        android:id="@+id/tv_2"
        android:textSize="30sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="20"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintHorizontal_bias="0.215"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="@id/tv_1"
        app:layout_constraintVertical_bias="0.078" />
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/tv_2"
        android:id="@+id/button"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:hint="Send Data TO Other Activity"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/tv_activity2"
        android:textSize="20sp"
        android:typeface="serif"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintTop_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical11;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import com.kuldeepjhala.practical11.databinding.ActivityMainBinding;
public class MainActivity extends AppCompatActivity {
```

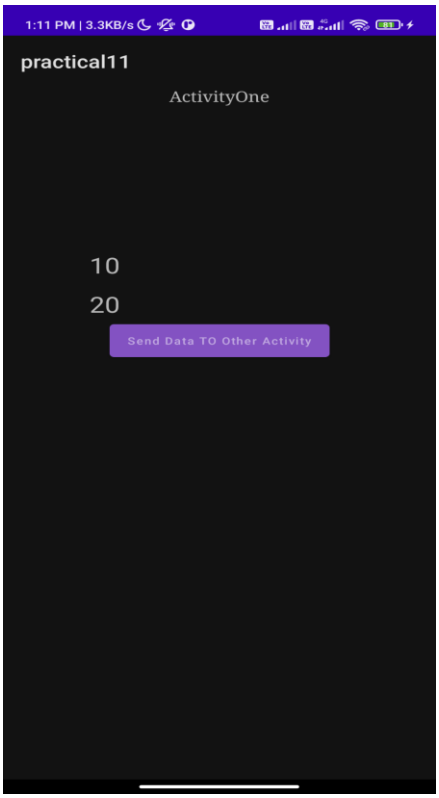
```
    ActivityMainBinding activityMainBinding;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        activityMainBinding = ActivityMainBinding.inflate(getLayoutInflater());
        setContentView(activityMainBinding.getRoot());
        TextView t1 = activityMainBinding.tv1;
        TextView t2 = activityMainBinding.tv2;
        activityMainBinding.button.setOnClickListener( l->{
                Intent i = new Intent(getApplicationContext(), MainActivity2.class);
                i.putExtra("Value1",Integer.parseInt(String.valueOf(t1.getText())));
                i.putExtra("Value2",Integer.parseInt(String.valueOf(t2.getText())));
                startActivity(i);
            }
        );
    }
}
```

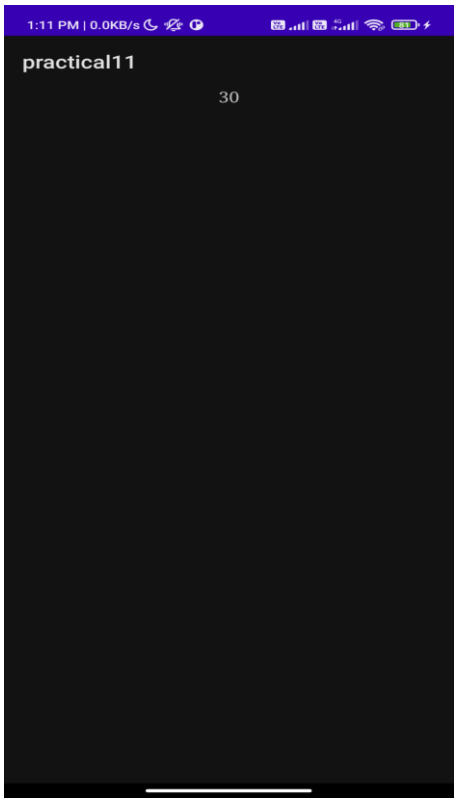
MainActivity2.java

```
package com.kuldeepjhala.practical11;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.TextView;
import com.kuldeepjhala.practical11.databinding.ActivityMain2Binding;
public class MainActivity2 extends AppCompatActivity {
    ActivityMain2Binding activityMain2Binding;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        activityMain2Binding = ActivityMain2Binding.inflate(getLayoutInflater());
        setContentView(activityMain2Binding.getRoot());
        TextView sum =  activityMain2Binding.tvActivity2;
        Bundle extras = getIntent().getExtras();
        int value1 = extras.getInt("Value1");
        int value2 = extras.getInt("Value2");
        int valueFinal = value1+value2;
        sum.setText(String.valueOf(valueFinal));
    }
}
```

Output





Practical 12

Aim : Create spinner with strings taken from resource folder(res >> value folder). On changing spinner value, change background of screen.

Activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical">
    <Spinner
        android:id="@+id/spinner1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="30dp"
        android:entries="@array/myarry"></Spinner>
    <ImageView
        android:id="@+id/imageView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:src="@drawable/image1"></ImageView>
</LinearLayout>
```

strings.xml

```
<resources>
    <string name="app_name">practical12</string>
    <string-array name="myarry">
        <item>January</item>
        <item>February</item>
        <item>March</item>
        <item>April</item>
```

```
    </string-array>
</resources>
```

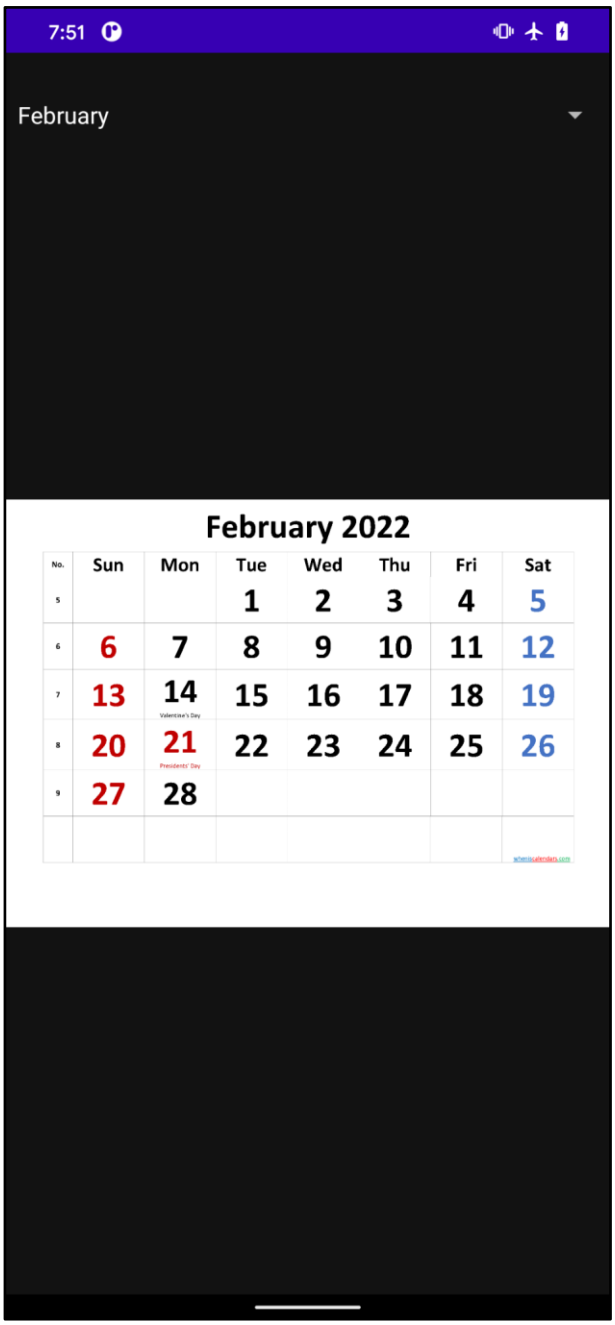
MainActivity.java

```
package com.kuldeepjhala.practical12;
import android.app.Activity;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ImageView;
import android.widget.Spinner;
public class MainActivity extends Activity {
    Spinner sp1;
    ImageView img1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        sp1 = (Spinner) findViewById(R.id.spinner1);
        img1 = (ImageView) findViewById(R.id.imageView1);
        sp1.setOnItemSelectedListener(new
                                AdapterView.OnItemSelectedListener() {
                                    @Override
                                    public void onItemSelected(AdapterView<?> arg0, View
                                            arg1,int arg2, long arg3) {
                                        // TODO Auto-generated method stub
                                        if (sp1.getSelectedItemId() == 0) {
```

```
                    img1.setImageResource(R.drawable.image1);
                } else if (sp1.getSelectedItemId() == 1) {
                    img1.setImageResource(R.drawable.image2);
                } else if (sp1.getSelectedItemId() == 2) {
                    img1.setImageResource(R.drawable.image3);
                } else if (sp1.getSelectedItemId() == 3) {
                    img1.setImageResource(R.drawable.image3);
                }
            }
            @Override
            public void onNothingSelected(AdapterView<?> parent) {
                // TODO Auto-generated method stub
            }
        });
    }
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        return super.onOptionsItemSelected(item);
    }
}
```

Output

7:50
January
January
2022
Sun Mon Tue Wed Thu Fri Sat
1
New Year's Day
2 3 4 5 6 7 8
9 10 11 12 13 14 15
16 17 18 19 20 21 22
Martin Luther King Day
23 24 25 26 27 28 29
30 31


7:51
February
February 2022
No. Sun Mon Tue Wed Thu Fri Sat
5 1 2 3 4 5
6 6 7 8 9 10 11 12
7 13 14 15 16 17 18 19
Valentine's Day
8 20 21 22 23 24 25 26
Presidents' Day
9 27 28

Practical 13

Aim :Create an application that will get the Text Entered in Edit Text and display that Text using toast (Message).

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".MainActivity">

    <TextView

        android:id="@+id/tv_mesage"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Message: "

        app:layout_constraintBottom_toBottomOf="parent"

        app:layout_constraintHorizontal_bias="0.244"

        app:layout_constraintLeft_toLeftOf="parent"

        app:layout_constraintRight_toRightOf="parent"

        app:layout_constraintTop_toTopOf="parent" />

    <EditText

        android:id="@+id/et_toast"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:hint="Enter the Message Here"

        app:layout_constraintBottom_toBottomOf="parent"
```

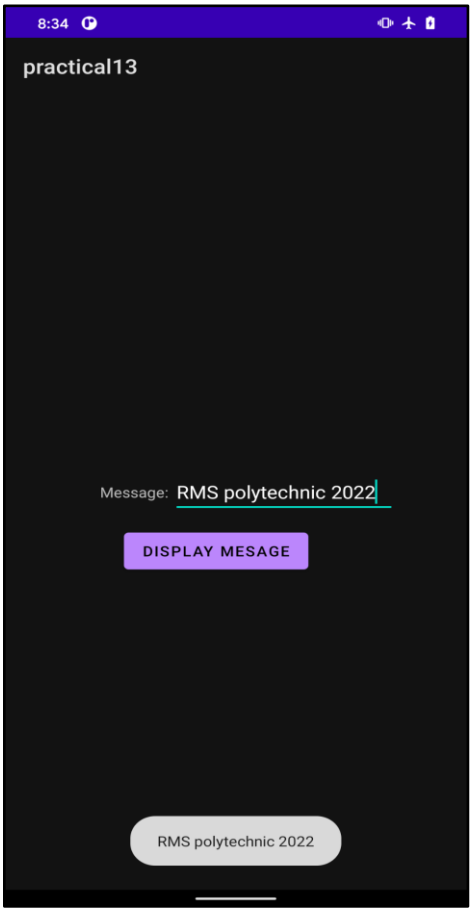
```
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toEndOf="@id/tv_mesage"
        app:layout_constraintTop_toTopOf="parent" />
    <Button
        android:id="@+id/bt_display"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Display Mesage"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.429"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/tv_mesage"
        app:layout_constraintVertical_bias="0.074" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical13;
import android.os.Bundle;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import com.kuldeepjhala.practical13.databinding.ActivityMainBinding;
public class MainActivity extends AppCompatActivity {
    ActivityMainBinding activityMainBinding;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        activityMainBinding = ActivityMainBinding.inflate(getLayoutInflater());

        setContentView(activityMainBinding.getRoot());

        EditText str = activityMainBinding.etToast;

        activityMainBinding.btDisplay.setOnClickListener(l -> {

            Toast toast = Toast.makeText(getApplicationContext(), str.getText().toString(),
Toast.LENGTH_SHORT);

            toast.show();

        });

    }

}
```

Output



Practical 14

Aim :Create an application that will Demonstrate Button onClick() Event and change the TextView Color based on button Clicked

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    tools:context=".MainActivity">

    <Button

        android:layout_width="wrap_content"

        android:id="@+id/btn_1"

        android:layout_height="wrap_content"

        app:layout_constraintTop_toTopOf="parent"

        app:layout_constraintBottom_toTopOf="@id/btn_2"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintEnd_toEndOf="parent"

        android:text="Yellow Color" />

    <TextView

        android:id="@+id/tv"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:textSize="20sp"

        android:text="Color of this text will be changed"

        app:layout_constraintTop_toBottomOf="@id/btn_1"
```

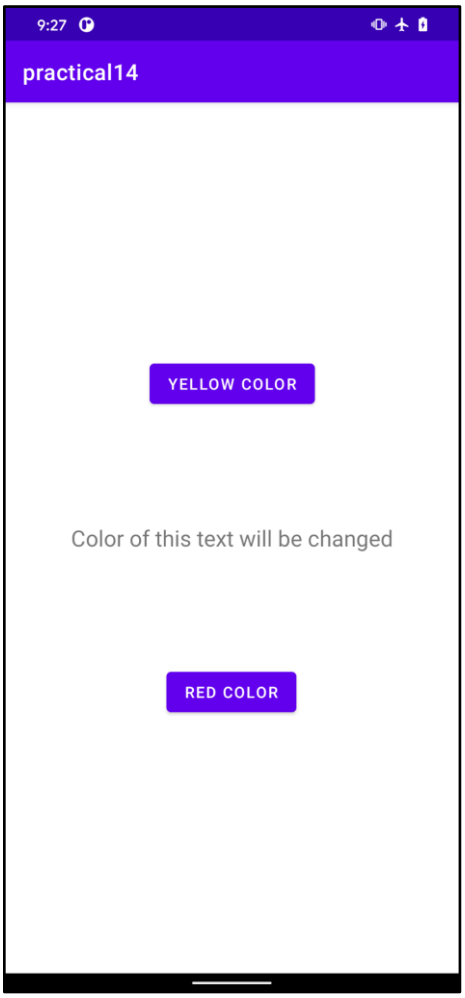
```
        app:layout_constraintBottom_toTopOf="@id/btn_2"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>
    <Button
        android:id="@+id/btn_2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Red Color"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/btn_1" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical14;
import androidx.appcompat.app.AppCompatActivity;
import android.annotation.SuppressLint;
import android.os.Bundle;
import com.kuldeepjhala.practical14.databinding.ActivityMainBinding;
public class MainActivity extends AppCompatActivity {
    ActivityMainBinding activityMainBinding;
    @SuppressLint("ResourceAsColor")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        activityMainBinding = ActivityMainBinding.inflate(getLayoutInflater());
        setContentView(activityMainBinding.getRoot());
```

```
        activityMainBinding.btn1.setOnClickListener(l->{
            activityMainBinding.tv.setTextColor(R.color.yellow);
        });
        activityMainBinding.btn2.setOnClickListener(l->{
            activityMainBinding.tv.setTextColor(R.color.red);
        });
    }
}
```

Output



Practical 15

Aim :Create an UI such that, one screen have list of all the types of cars. On selecting of any car name, next screen should show Car details like: name, launched date, company name

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".MainActivity">

    <TextView

        android:id="@+id/tv_heading"

        style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:layout_marginTop="92dp"

        android:text="CARS"

        android:textSize="30sp"

        android:textAlignment="center"

        app:layout_constraintHorizontal_bias="0.498"

        app:layout_constraintLeft_toLeftOf="parent"

        app:layout_constraintRight_toRightOf="parent"

        app:layout_constraintTop_toTopOf="parent" />

    <ListView

        android:id="@+id/lv_Main"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/tv_heading"
        app:layout_constraintVertical_bias="0.025">
    </ListView>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical15;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import com.kuldeepjhala.practical15.databinding.ActivityMainBinding;
public class MainActivity extends AppCompatActivity {
    ActivityMainBinding activityMainBinding;
    String cars[]={
      "Verna","Elentra","Civic"
    };
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        activityMainBinding = ActivityMainBinding.inflate(getLayoutInflater());
        setContentView(activityMainBinding.getRoot());
        ListView carslist = activityMainBinding.lvMain;
        ArrayAdapter<String> arr;
            arr = new ArrayAdapter<String>(this,
R.layout.support_simple_spinner_dropdown_item,
            cars);
        carslist.setAdapter(arr);
        carslist.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> a, View v, int position,
                                    long id) {
                if(position == 0){
                    Intent intent = new Intent(MainActivity.this, MainActivity2.class);
                    startActivity(intent);
                } else if(position == 1)
                {
                    Intent intent2 = new Intent(MainActivity.this, MainActivity3.class);
                    startActivity(intent2);
                } else if(position == 2){
                    Intent intent3 = new Intent(MainActivity.this, MainActivity4.class);
                    startActivity(intent3);
                }
            }
        });
    }
}
```

Activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">
    <TextView
        android:id="@+id/heading"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Verna"
        android:textSize="40sp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox"
        />
    <ImageView
        app:layout_constraintTop_toBottomOf="@id/heading"
        android:id="@+id/img_car"
        app:layout_constraintStart_toStartOf="parent"
        android:layout_width="415dp"
        android:layout_height="330dp"
        android:src="@drawable/verna"
        />
    <TextView
        android:id="@+id/tv_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
        app:layout_constraintTop_toBottomOf="@id/img_car"
        android:text="Name:Verna"/>
    <TextView
        android:id="@+id/tv_date"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/tv_name"
        android:text="Launch Date:2021 January"/>
    <TextView
        android:id="@+id/tv_company_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/tv_date"
        android:text="Company Name: Hyundai"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Activity_main3.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity3">
    <TextView
        android:id="@+id/heading"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Elentra"
```

```
        android:textSize="40sp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox"
        />
    <ImageView
        app:layout_constraintTop_toBottomOf="@id/heading"
        android:id="@+id/img_car"
        app:layout_constraintStart_toStartOf="parent"
        android:layout_width="415dp"
        android:layout_height="330dp"
        android:src="@drawable/elentra"
        />
    <TextView
        android:id="@+id/tv_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/img_car"
        android:text="Name:elentra"/>
    <TextView
        android:id="@+id/tv_date"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/tv_name"
        android:text="Launch Date:2021 March"/>
    <TextView
        android:id="@+id/tv_company_name"
        android:layout_width="match_parent"
```

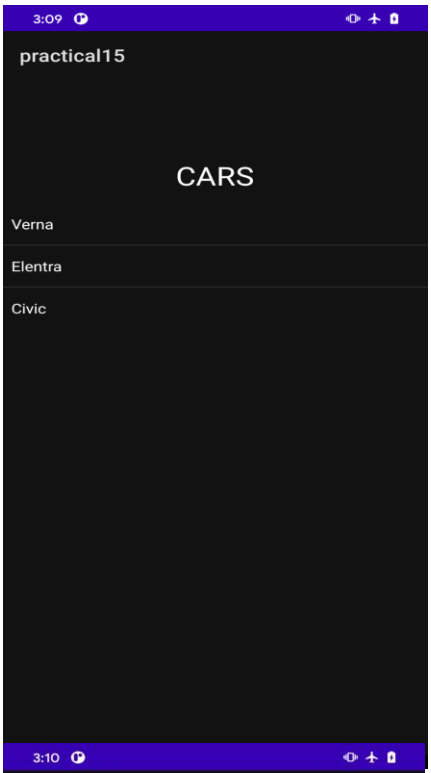
```
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/tv_date"
        android:text="Company Name: Hyundai"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

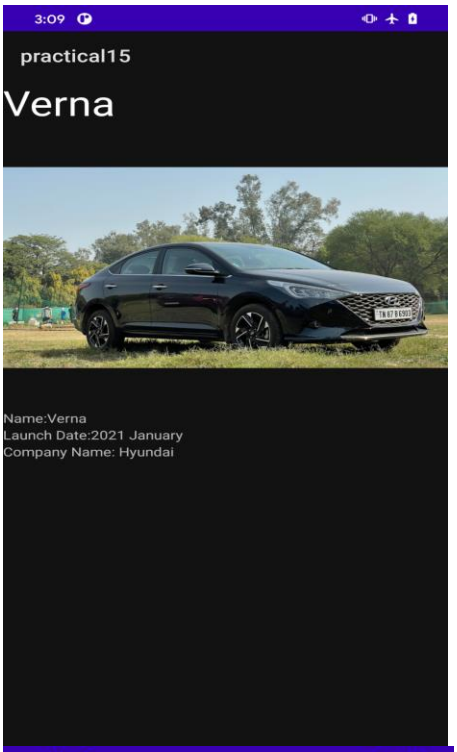
Activity_main4.xml

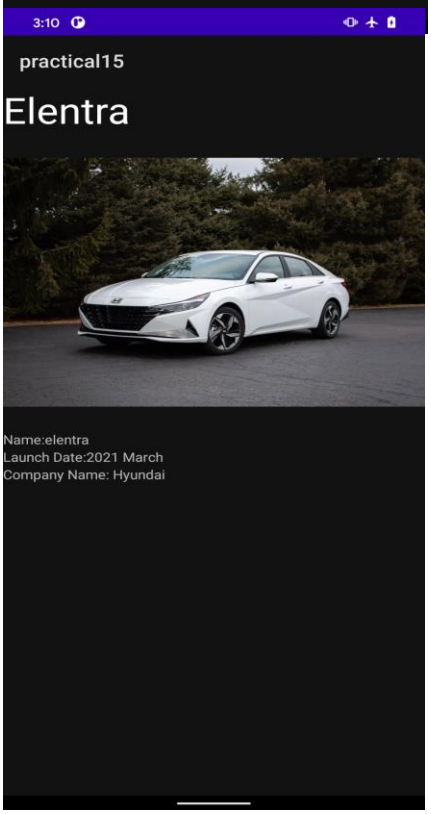
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity4">
    <TextView
        android:id="@+id/heading"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Civic"
        android:textSize="40sp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox"
        />
    <ImageView
        app:layout_constraintTop_toBottomOf="@id/heading"
        android:id="@+id/img_car"
        app:layout_constraintStart_toStartOf="parent"
        android:layout_width="415dp"
        android:layout_height="330dp"
```

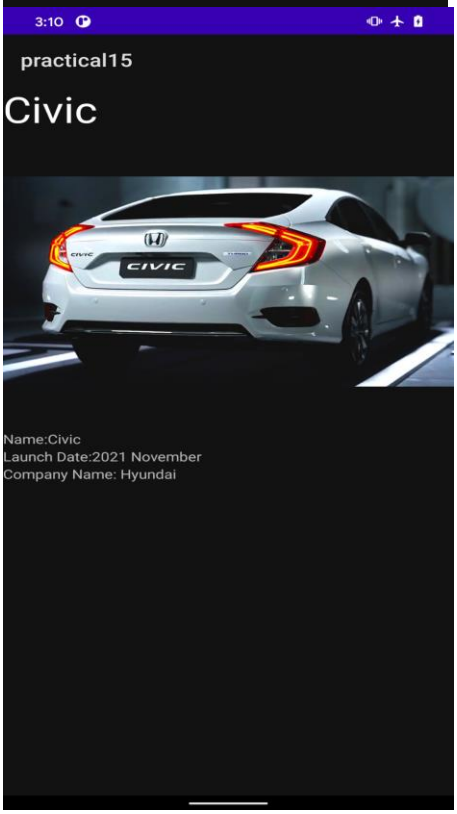
```
        android:src="@drawable/civic"
        />
    <TextView
        android:id="@+id/tv_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/img_car"
        android:text="Name:Civic"/>
    <TextView
        android:id="@+id/tv_date"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/tv_name"
        android:text="Launch Date:2021 November"/>
    <TextView
        android:id="@+id/tv_company_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toBottomOf="@id/tv_date"
        android:text="Company Name: Hyundai"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output


3:09
practical15
CARS
Verna
Elentra
Civic


3:09
practical15
Verna
Name:Verna
Launch Date:2021 January
Company Name: Hyundai


3:10
practical15
Elentra
Name:elentra
Launch Date:2021 March
Company Name: Hyundai


3:10
practical15
Civic
CIVIC
Name:Civic
Launch Date:2021 November
Company Name: Hyundai

Practical 16

Aim :Create an application that will Demonstrate Dialog Box Control In Android

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical">
    <Button
        android:id="@+id/buttonShowCustomDialog"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Custom Dialog" />
</LinearLayout>
```

Custom.xml

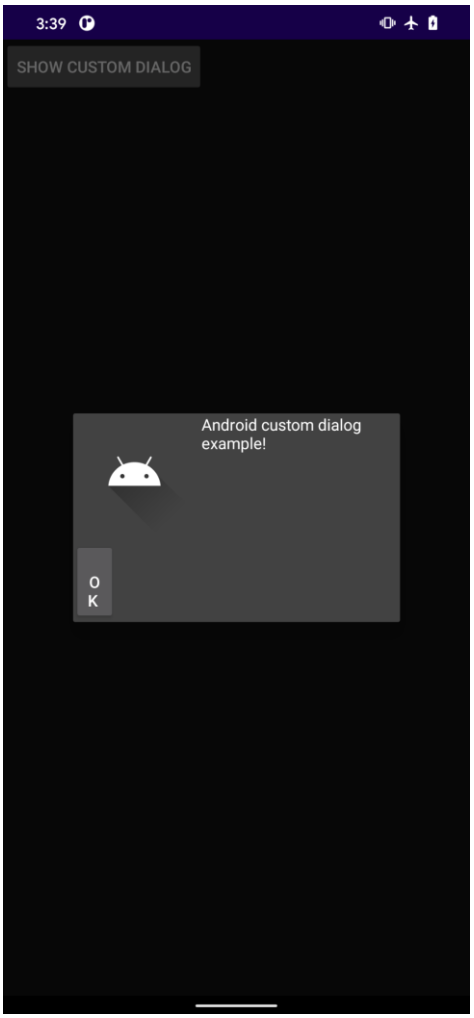
```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent">
    <ImageView
        android:id="@+id/image"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginRight="5dp" />
    <TextView
        android:id="@+id/text"
        android:layout_width="fill_parent"
```

```
        android:layout_height="wrap_content"
        android:layout_toRightOf="@+id/image"
        android:textColor="#FFF" />/>
    <Button
        android:id="@+id/dialogButtonOK"
        android:layout_width="100px"
        android:layout_height="wrap_content"
        android:layout_below="@+id/image"
        android:layout_marginTop="5dp"
        android:layout_marginRight="5dp"
        android:text=" Ok " />
</RelativeLayout>
```

MainActivity.java

```
package com.kuldeepjhala.practical16;
import android.app.Activity;
import android.app.Dialog;
import android.content.Context;
import android.os.Bundle;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
public class MainActivity extends Activity {
    final Context context = this;
    private Button button;
    public void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
setContentView(R.layout.activity_main);
button = (Button) findViewById(R.id.buttonShowCustomDialog);
// add button listener
button.setOnClickListener(new OnClickListener() {
    @Override
    public void onClick(View arg0) {
        // custom dialog
        final Dialog dialog = new Dialog(context);
        dialog.setContentView(R.layout.custom);
        dialog.setTitle("Title...");
        // set the custom dialog components - text, image and button
        TextView text = (TextView) dialog.findViewById(R.id.text);
        text.setText("Android custom dialog example!");
        ImageView image = (ImageView)
                dialog.findViewById(R.id.image);
        image.setImageResource(R.drawable.ic_launcher_foreground);
        Button dialogButton = (Button)
                dialog.findViewById(R.id.dialogButtonOK);
        // if button is clicked, close the custom dialog
        dialogButton.setOnClickListener(new OnClickListener() {
            @Override
            public void onClick(View v) {
                dialog.dismiss();
            }
        });
        dialog.show();
    }
});
```

```
    }
}
```

Output